

NB14 v2: Stacking Ensemble (5-Model Diversity)

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Degisiklik Ozeti (v2)

A1) Meta probability uzerinde threshold optimizasyonu (F1-max).

A2) NB12 XGBoost+OHE baseline yeniden uretildi: F1=0.8877, AUC=0.8388, thr=0.310. NB12 referans: F1=0.9020 AUC=0.8404.

A3) Sabit sutunlar ve birebir ayni sutun ciftleri (CAT_3/CAT_5 dahil) train sonrasi drop edildi.

B1) Taban modeller 7 -> 5 cesitliliği: CatBoost (native cat), LightGBM (LE), Logistic L1, TabNet, kNN.

B3) OOF probability uzerinde IsotonicRegression; meta-feature matrisi kalibre olasiliklardan kuruluyor.

Balance B2 duzeltildi: B1 ile ayni test seti, train icinde patho. undersample, fazla patho. DROP (test'e tasinmaz, ID overlap=0).

1. Sonuclar (MASTER Hold-Out)

Config	Meta	Thr	Tr-F1	Te-F1	Te-Prec	Te-Rec	Te-AUC	Te-MCC	N-feat
B1_M1	LR	0.19	0.8922	0.8942	0.8419	0.9535	0.8636	0.5452	287
B1_M1	LightGBM	0.25	0.9033	0.8935	0.8462	0.9465	0.8484	0.5468	287
B1_M3	LR	0.20	0.8944	0.8931	0.8491	0.9419	0.8636	0.5482	309
B1_M3	LightGBM	0.28	0.9094	0.8906	0.8562	0.9279	0.8504	0.5484	309
B1_M5	LR	0.17	0.8935	0.8938	0.8447	0.9488	0.8668	0.5462	172
B1_M5	LightGBM	0.30	0.9054	0.8741	0.8762	0.8721	0.8504	0.5324	172
B2_M1	LR	0.38	0.7894	0.8738	0.8779	0.8698	0.8525	0.5343	287
B2_M1	LightGBM	0.40	0.8297	0.8680	0.8881	0.8488	0.8325	0.5370	287
B2_M3	LR	0.35	0.7931	0.8803	0.8635	0.8977	0.8561	0.5288	309
B2_M3	LightGBM	0.39	0.8271	0.8838	0.8747	0.8930	0.8511	0.5532	309
B2_M5	LR	0.43	0.7923	0.8592	0.8985	0.8233	0.8570	0.5356	172
B2_M5	LightGBM	0.33	0.8268	0.8793	0.8616	0.8977	0.8543	0.5235	172

2. En İyi Konfigurasyon: B1_M1

Model	Test F1 (calib)	Test AUC	OOF F1 (calib)
catboost	0.8966	0.8521	0.8870
lightgbm	0.8945	0.8321	0.8813
logreg_l1	0.8779	0.8227	0.8741
mlp3	0.8733	0.7872	0.8609
deep_mlp	0.8680	0.7650	0.8586
knn	0.8366	0.7404	0.8379

Stacking LR (calibrated + threshold-tuned) test F1: 0.8942 thr=0.19

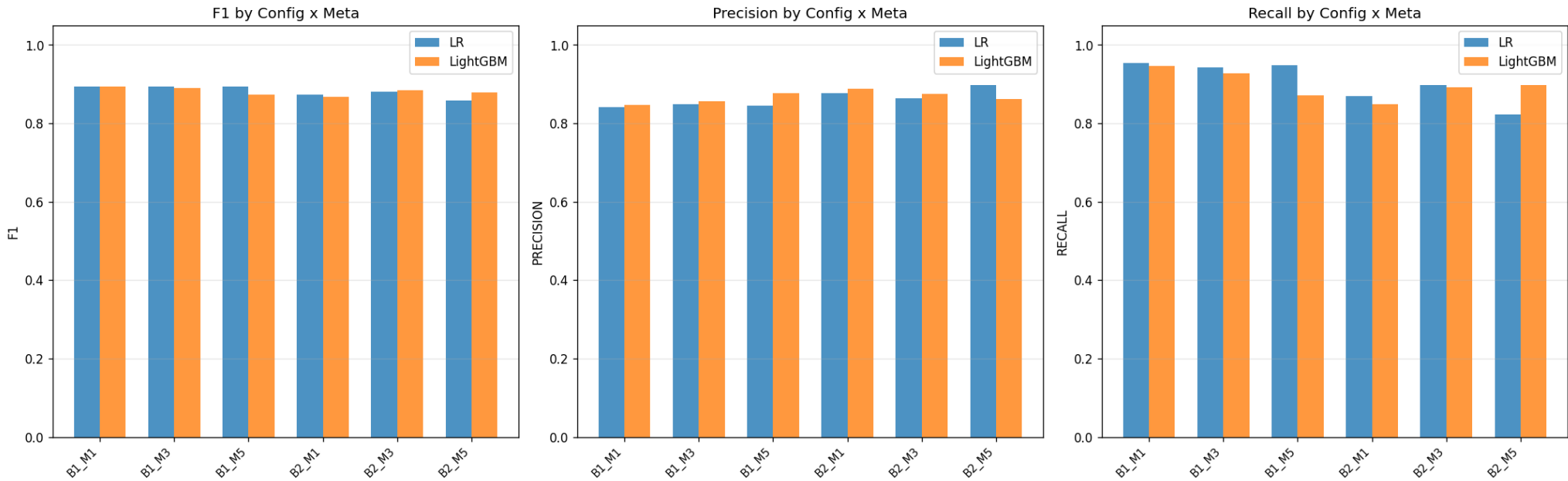
A3 redundant drops at best config: ['AL_312', 'AL_164', 'AL_208', 'AL_182', 'AL_284', 'AL_263', 'AL_191', 'AL_267', 'AL_128', 'AL_179', 'AL_131', 'AL_299', 'AL_276', 'CAT_5', 'AL_116', 'AL_320', 'AL_303', 'AL_170', 'AL_332', 'AL_309', 'AL_152', 'AL_167', 'AL_272', 'AL_212', 'AL_195', 'AL_143', 'AL_80', 'AL_113', 'AL_280', 'AL_146', 'AL_307', 'AL_248', 'AL_197', 'AL_119', 'AL_176', 'AL_292', 'AL_252', 'AL_158', 'AL_240', 'AL_185', 'AL_140', 'AL_220', 'AL_227', 'AL_161', 'AL_125', 'AL_173', 'AL_269', 'AL_324', 'AL_233', 'AL_101', 'AL_155', 'AL_134', 'AL_104', 'AL_122', 'AL_244', 'AL_137', 'AL_256', 'AL_107', 'AL_204', 'AL_236', 'AL_231', 'AL_110', 'AL_316', 'AL_149']

3. Panel Transfer Sonuclari (Top-5 by MASTER F1)

Config	Panel	Thr	F1	Recall	Precision	AUC	MCC
B1_M1	CFTR	0.19	0.9451	0.9556	0.9348	0.9344	0.6965
B1_M1	KANSER	0.19	0.8983	0.9888	0.8230	0.9299	0.6321
B1_M1	PAH	0.19	0.9329	0.9871	0.8844	0.8337	0.4998
B1_M3	CFTR	0.20	0.9438	0.9333	0.9545	0.9423	0.7178
B1_M3	KANSER	0.20	0.9085	0.9813	0.8457	0.9321	0.6737
B1_M3	PAH	0.20	0.9290	0.9710	0.8905	0.8362	0.4839
B1_M5	CFTR	0.17	0.9508	0.9667	0.9355	0.9339	0.7236
B1_M5	KANSER	0.17	0.9010	0.9851	0.8302	0.9333	0.6431
B1_M5	PAH	0.17	0.9313	0.9839	0.8841	0.8365	0.4865
B2_M1	CFTR	0.38	0.8876	0.8333	0.9494	0.9270	0.5558
B2_M1	KANSER	0.38	0.9097	0.9403	0.8811	0.9198	0.6898
B2_M1	PAH	0.38	0.9250	0.9548	0.8970	0.8311	0.4786
B2_M3	CFTR	0.35	0.9266	0.9111	0.9425	0.9323	0.6403
B2_M3	KANSER	0.35	0.9117	0.9627	0.8658	0.9287	0.6892
B2_M3	PAH	0.35	0.9286	0.9645	0.8952	0.8444	0.4922

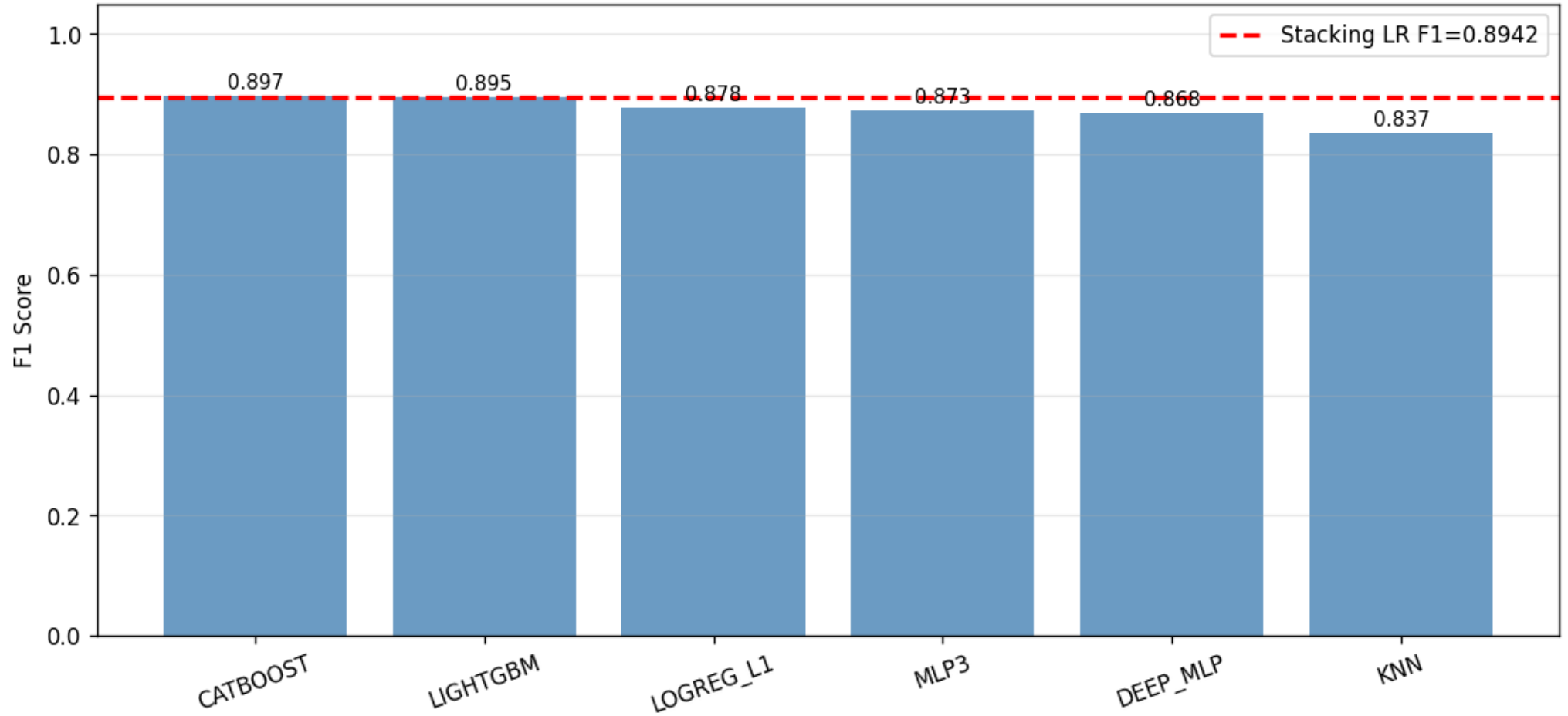
4. MASTER F1 / Precision / Recall

MASTER Hold-Out Results (calibrated probs, threshold-tuned meta)



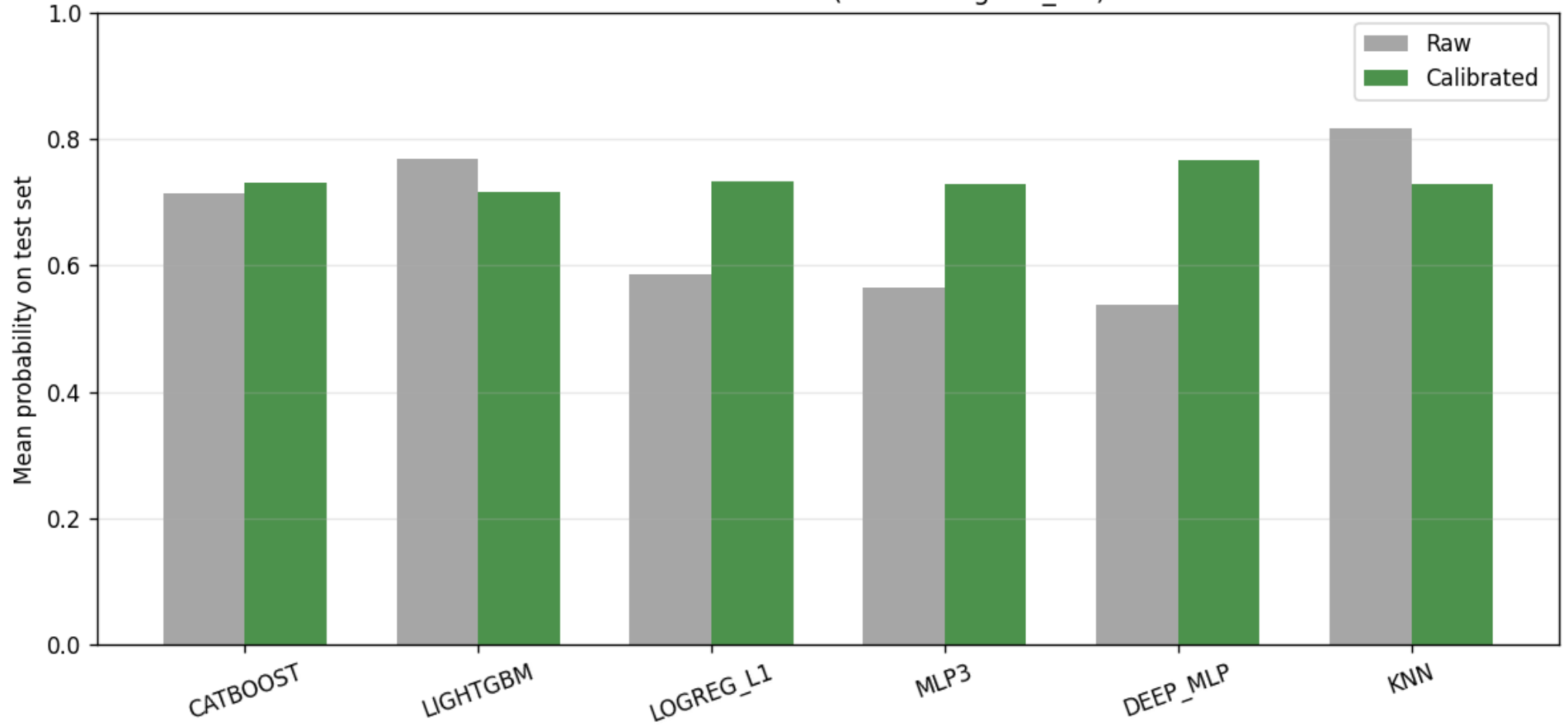
5. Taban Modeller vs Stacking

Base Model Test F1 vs Stacking (B1_M1)



6. B3 Calibration Etkisi

B3 Calibration Effect (best config: B1_M1)



7. Konfuzyon Matrisleri

